

Gene Expression Analysis with the GE 24.192 IFC Using TaqMan Assays

Contents

About This Guide	2	Appendix A: Available Bundle Kits	17
Safety Alert Conventions	2	Appendix B: Reagent Kit Components	18
Safety Data Sheets	3	Appendix C: Biomark HD and Biomark Thermal Cycler Protocol	18
Introduction	3	Appendix D: Related Documents	19
Workflow Overview from RNA	3	Appendix E: Safety	19
Materials	4	General Safety	19
Kits and Reagents	4	Instrument Safety	19
Required Equipment	5	Chemical Safety	20
Sample Requirements	5	Disposal of Products	20
Required Software	6		
Before You Begin	6		
Prepare the cDNA	6		
Retrieve Reagents	6		
Prepare RT Reactions	7		
Preamplify the cDNA	8		
Retrieve the Reagents	8		
Prepare the Preamplification Reactions	8		
Prepare Assays	9		
Retrieve Reagents	9		
Prepare 2X Assays	10		
Prepare the Final Assay Mix	11		
Dilute the Preamplified cDNA and Prepare Final Sample Mixes	12		
Load the IFC	12		
Collect Real-Time PCR Data	15		
Analyze the Data	16		

About This Guide

This protocol describes how to perform gene expression analysis using the 24.192 Dynamic Array™ IFC (integrated fluidic circuit) for gene expression and the Biomark™ HD or Biomark system. For detailed instructions on instrument and software operation, see the Biomark HD Data Collection User Guide (PN 100-2451) or the Biomark/EP1™ Data Collection User Guide (PN 68000127).

IMPORTANT Before using this protocol, read and understand the detailed instructions and safety guidelines in this document. For complete safety information, see [Appendix E](#).

Safety Alert Conventions

Fluidigm documentation uses specific conventions for presenting information that may require your attention. Refer to the following safety alert conventions.

Safety Alerts for Chemicals

For hazards associated with chemicals, this document follows the United Nations Globally Harmonized System of Classification and Labelling of Chemicals (GHS) and uses indicators that include a pictogram and a signal word that indicates the severity level:

Indicator	Description
	Pictogram (see example) consisting of a symbol on a white background within a red diamond-shaped frame. Refer to the individual safety data sheet (SDS) for the applicable pictograms and hazards pertaining to the chemicals being used.
DANGER	Signal word that indicates more severe hazards.
WARNING	Signal word that indicates less severe hazards.

Safety Alerts for Instruments

For hazards associated with instruments, this document uses indicators that include a pictogram and signal words that indicate the severity level:

Indicator	Description
	Pictogram (see example) consisting of a symbol on a white background within a black triangle-shaped frame. Refer to the instrument user guide for the applicable pictograms and hazards pertaining to instrument usage.
DANGER	Signal word that indicates an imminent hazard that will result in severe injury or death if not avoided.
WARNING	Signal word that indicates a potentially hazardous situation that could result in serious injury or death if not avoided.
CAUTION	Signal word that indicates a potentially hazardous situation that could result in minor or moderate personal injury if not avoided.
IMPORTANT	Signal word that indicates information necessary for proper use of products or successful outcome of experiments.

Safety Data Sheets

Read and understand the SDSs before handling chemicals. To obtain SDSs for chemicals ordered from Fluidigm, either alone or as part of this system, go to fluidigm.com/sds and search for the SDS using either the product name or the part number.

Some chemicals referred to in this user guide may not have been provided with your system. Obtain the SDSs for chemicals provided by other manufacturers from those manufacturers.

Introduction

This protocol is intended to be used for gene expression analysis on the Biomark™ HD and Biomark system using the 24.192 Dynamic Array IFC for gene expression and Applied Biosystems® TaqMan® probes and primers. The GE 24.192 IFC uses 24 samples and 192 assays at the same time for a single IFC run, which enables 4,608 real time PCR reactions simultaneously.

Workflow Overview from RNA

We recommend preparing the assays and sample dilution buffer while the preamplification reaction is running.

Workflow Step		Hands-On Time	Run Time	Total Time
1 Prepare the cDNA.	Potential stopping point	15 min	40 min	55 min
Reverse transcription reaction				
2 Preamplify the cDNA.	Potential stopping point	15 min	1 h 20 min*	1 h 35 min
3 Prepare assays and sample dilution buffer.		20 min	—	
Prepare the assays and sample dilution buffer while the preamplification reaction is running.				
4 Dilute samples		5 min	—	5 min
5 Load the IFC.	Prime, Load and Mix (one step)	5 min	30 min	35 min
Juno™ or IFC Controller RX				
6 Thermal-cycle and collect data.		—	1 h 30 min	1 h 35 min
Biomark HD or Biomark)				
Total		1 h	4 h	5 h

* Estimated for 16 cycles

Materials

Kits and Reagents

Fluidigm provides two configurations of bundled kits: one that includes the Preamp and Reverse Transcription Master Mixes for cDNA preparation and preamplification using the Fluidigm procedures, and one without the master mixes. Both configurations are available in 2-IFC and 10-IFC bundles.

This protocol was optimized using the TaqMan™ Gene Expression Master Mix (Thermo Fisher Scientific PN 4369016). Other master mixes may be used successfully, such as the TaqMan™ Universal PCR Master Mix (Thermo Fisher Scientific PN 4304437) and TaqMan™ Fast Advanced Master Mix (Thermo Fisher Scientific PN 4444557). For more information on using alternate master mixes, contact Fluidigm technical support.

Required Kit from Fluidigm

This protocol describes using the Biomark 24.192 Complete Bundle—Probe-Based GE kit. For a list of all the available bundled kit configurations, see [Appendix A](#). For a list of the reagent kit components, see [Appendix B](#).

IMPORTANT Store reagents as soon as they are received, according to manufacturer's storage recommendations.

Product Name	Source	Part Number	Storage
Biomark 24.192 Complete Bundle—Probe-Based GE	Fluidigm	101-7659 (2 IFCs)	
Kit contains:		101-7660 (10 IFCs)	
• Biomark 24.192 GE Reagent Kit			–20 °C
• Preamp and Reverse Transcription Master Mix Kit*			–20 °C
• Biomark Control Line Fluid 192.24 & 24.192			Room temperature
• 24.192 Dynamic Array IFC for Gene Expression			Room temperature

* The Preamp and Reverse Transcription Master Mix kit is included in the complete bundle, and is required if you are using the Fluidigm procedures for cDNA preparation and preamplification. For additional configurations of the Preamp and Reverse Transcription Master Mix reagent kits, go to fluidigm.com.

Required Reagents from Other Suppliers

Product Name	Source	Part Number
TaqMan® Gene Expression Assays (20X) (up to 192 assays) Concentration: primers, 18 µM; probe, 5 µM.	Thermo Fisher Scientific	—
TaqMan™ Gene Expression Master Mix	Thermo Fisher Scientific	4369016
DNA suspension buffer (10 mM Tris, 0.1 mM EDTA, pH 8.0)*	Teknova	T0223 (250 mL)
PCR-grade water	—	—

* Recommended if Teknova is not available in your location: 1X TE Buffer (Thermo Fisher Scientific PN 12090015 [100 mL]).

Required Consumables

Product	Source	Part Number
5.0 mL tubes	Major laboratory supplier (MLS)	—
25 mL reagent reservoir [†]	MLS	—
1.5 mL microtubes	MLS	—
0.2 mL PCR tubes	MLS	—
PCR-compatible adhesive seals*	MLS	—
96-well PCR plates	MLS	—
8-well PCR tube strips	MLS	—
Filtered pipette tips	MLS	—
Clear tape	MLS	—

* Recommended: MicroAmp® Clear Adhesive Film (Thermo Fisher Scientific PN 4306311).

[†] Recommended: Accuflo 25-mL Reagent Reservoir, Certified Sterile (Accuflo PN D620025)

Required Equipment

Product	Source	Part Number
Biomark HD system or Biomark	Fluidigm	BMKHD-BMKHD or BMK-BMK
Juno system with RX Interface Plate or IFC Controller RX	Fluidigm	101-6455 w/100-6822 or IFC-RX
Two centrifuges: one for microtubes, one for 96-well PCR plates	MLS	—
Pipettes (single channel and 8-channel)*	MLS	—
Standard thermal cycler	MLS	—
DNA-free hood	MLS	—
Vortexer	MLS	—

* Recommended: Rainin® pipettes

Sample Requirements

This protocol supports using cDNA that is prepared using original RNA samples in a range of 2–200 ng of total RNA for each reverse transcription (RT) reaction. Success of a given sample for relative quantification using qPCR will depend on both the abundance and the quality of expressed mRNA transcripts within the sample. Adding lower amounts of total RNA or RNA of lower quality may result in reduced detection of low-expressed mRNA transcripts.

We recommend running control samples for each sample set.

For better traceability, assemble cDNA samples in a 96-well PCR plate and record in a sample map.

Required Software

Fluidigm Real-Time PCR Analysis software v4.5.1 or later and Biomark Data Collection software v4.5.1 or later is required for this protocol. For software updates, go to fluidigm.com/software.

Before You Begin

IMPORTANT Read and understand the safety information in [Appendix E](#).

For the overall success of the protocol, we recommend the following best practices.

- Protect assays from light and store in the refrigerator or freezer when not in use. Assays are light-sensitive.
- Ensure that lab consumables (tubes, tips, plates) used for the RNA handling steps are RNase-free.
- Thaw reagents at room temperature (15–30 °C) unless directed to thaw them on ice.
- Mix and centrifuge reagents as directed.
- Avoid creating bubbles when transferring reagents to the IFC.
 - Check the source plate or tube for bubbles before pipetting and pipette tips for air gaps while pipetting.
 - Pipet into the IFC inlets at an angle and always stop at the first stop on the pipette to avoid creating bubbles in the inlets. If a bubble is introduced, ensure that it floats to the top of the inlet.
 - If necessary, remove any bubbles from an IFC inlet by removing the contents of the inlet by pipette and then carefully re-pipetting the contents into the inlet.

To distribute same mix into 24 or 192 wells, we recommend first aliquoting reagents into an 8-well PCR tube strip to enable transfer into a 96-well PCR plate using an 8-channel pipette.

Prepare the cDNA

If you already have cDNA, skip this section and go to [Preamplify the cDNA on page 8](#), or follow your own protocol for Reverse Transcription reactions.

Retrieve Reagents

Required Reagent	Preparation
PCR Water (Fluidigm PN 100-5941)	 Remove from –20 °C, thaw to room temperature.
Reverse Transcription Master Mix (Fluidigm PN 100-6297)	
RNA (user-supplied), 2–200 ng/μL	Remove from –20 °C, thaw on ice, and keep on ice.

Prepare RT Reactions

- 1 Briefly vortex and centrifuge the reagents before using.
- 2 On ice, prepare the RT pre-mix in a 1.5 mL PCR tube as shown in Table 1.

Table 1. RT pre-mix

Component		Volume per Reaction (μL)	Volume for 24 Reactions (μL)*
Reverse Transcription Master Mix (Fluidigm PN 100-6297)		1.0	32
PCR Water (Fluidigm PN 100-5941)		3.0	96
Total		4.0	128

* Includes overage for ease of pipetting

- 3 Aliquot 14.5 μL of the RT pre-mix into each well of an 8-well PCR tube strip.
- 4 Prepare the RT sample mix as shown in Table 2.

Table 2. RT sample mix

Component	Volume per Reaction (μL)
RT pre-mix (see Table 1)	4.0
RNA, 2–200 ng/μL	1.0
Total	5.0

- a Using an 8-channel pipette, transfer RT pre-mix to individual wells of a 96-well PCR plate.
 - b Add RNA samples to each well containing RT pre-mix.
- 5 Properly seal the plate with a PCR-compatible adhesive seal and gently vortex to mix the RT reactions.
 - 6 Centrifuge the RT reactions and then place in a standard thermal cycler.
 - 7 Incubate the RT reactions using the following thermal protocol:

Temperature	Time	Cycle
25 °C	5 min	Hold
42 °C	30 min	Hold
85 °C	5 min	Hold
4 °C	∞	Hold

STOPPING POINT The reverse-transcription reaction products (cDNA) can be used immediately for preamplification reactions with Preamp Master Mix or stored at –20 °C for later use.

Preamplify the cDNA

If you are using your own procedure for preamplification, skip this section and go to [Prepare Assays on page 9](#).

Retrieve the Reagents

Required Reagent	Preparation
cDNA	
PCR Water (Fluidigm PN 100-5941)	
20X GE Sample Loading Reagent (Fluidigm PN 85000746)	
DNA suspension buffer (Teknova PN T0223)	
TaqMan Gene Expression Assays (20X)	Remove from –20 °C, thaw to room temperature. Protect assays from light while thawing.
Preamp Master Mix (Fluidigm PN 100-5744)*	 Remove from –20 °C, thaw to room temperature.

* Included in kit configurations PN 101-7659 and PN 101-7660

Prepare the Preamplification Reactions

- 1 Dilute the 20X assays with PCR Water to make a 0.1X preamp assay pool:

Table 3. Preamp assay pool

Component	Volume (μL)
20X assay (up to 192 assays)	2 x number of assays
PCR Water (Fluidigm PN 100-5941)	 400 – (2 x number of assays)
Total	400

- a Combine 2 μL of each 20X assay in a single 1.5 mL tube (up to 192 assays).
- b Add PCR water to bring the volume up to 400 μL. For example, if you are using 173 assays, combine 2 μL of each assay and 54 μL of PCR water.

- 2 Briefly vortex and centrifuge the pool for 10 seconds.
- 3 In a DNA-free hood, prepare the preamplification pre-mix in a 1.5 mL PCR tube as shown in [Table 4](#).

Table 4. Preamplification pre-mix

Component		Volume per Reaction (μL)	Volume for 24 Reactions (μL)*
0.1X Preamp Assay Pool (from Step 1)		2.5	80
Preamp Master Mix (Fluidigm PN 100-5744)		1.00	32
PCR Water (Fluidigm PN 100-5941)		0.25	8.0
Total		3.75	120

* Includes overage

- 4 Briefly vortex and centrifuge the pre-mix for 10 seconds.
- 5 Aliquot 13.0 μL of the preamplification pre-mix into each tube of an 8-well strip.
- 6 Prepare preamplification sample mix as shown in Table 5.

Table 5. Preamplification sample mix

Component	Volume per Reaction (μL)
Preamplification pre-mix (see Table 4)	3.75
cDNA	1.25
Total	5.00

- a Using an 8-channel pipette, transfer 3.75 μL preamplification pre-mix into individual wells of a 96-well PCR plate.
 - b Remove the plate from the DNA-free hood and add 1.25 μL cDNA samples to each well containing pre-mix.
- 7 Seal the plate using a PCR-compatible adhesive seal.
- 8 Mix the reactions by briefly vortexing and then centrifuge.
- 9 Place the plate in the thermal cycler and use the following thermal protocol:

Temperature	Time	Cycle	
95 °C	2 min	Hold	
95 °C	15 sec	Denaturation	16 cycles*
60 °C	4 min	Annealing/extension	
4 °C	∞	Hold	

* Recommended starting point (within a range of 10–20 cycles). The appropriate number of cycles may need to be determined empirically, as sample sets may vary with respect to sample amount and quality. For best performance, the target Cq range on GE 24.192 IFC should be between 8–22.

- 10 While preamplification is running, you can prepare the assay mixes in the next section and the sample dilution mix on page 12.

Prepare Assays

Retrieve Reagents

Required Reagent		Preparation
2X Assay Loading Reagent (Fluidigm PN 85000736)		If frozen, thaw to room temperature. For subsequent uses, keep at 4 °C.
20X GE Sample Loading Reagent (Fluidigm PN 85000746)		
TaqMan Gene Expression Assays (20X)		Remove from storage and keep at recommended temperature.
DNA Suspension Buffer (Teknova PN T0223)		
Gene Expression Master Mix (2X) (Thermo Fisher Scientific 4369016)		

Prepare 2X Assays

- 1 Be certain your 20X TaqMan assays have the following concentration: primers, 18 μM ; probe, 5 μM .
- 2 In a DNA-free hood, prepare a 2.22X assay dilution mix by combining the components shown in Table 6 in a new 5–15 mL tube:

Table 6. Assay dilution mix

Component		Volume (μL)*
2X Assay Loading Reagent (Fluidigm PN 85000736)	●	250
20X GE Sample Loading Reagent (Fluidigm PN 85000746)	●	250
DNA suspension buffer (Teknova PN T0223)		1,750
Total		2,250

* Includes overage

- 3 Vortex thoroughly or mix with a 1 mL pipette by pipetting up and down 10 times.
- 4 Aliquot 125 μL of assay dilution mix into each well of two 8-well strips (see Figure 1).
- 5 Using an 8-channel pipette, transfer 9 μL of assay dilution mix to each well of two new 96-well PCR plates.
- 6 Add 1 μL of 20X assay to each well. Seal the plates carefully. Briefly vortex and centrifuge the plates for 60 seconds.

IMPORTANT

- All wells must contain assay dilution mix to make the final assay mix.
- Add 1 μL PCR water to any wells that do not contain a 20X assay.

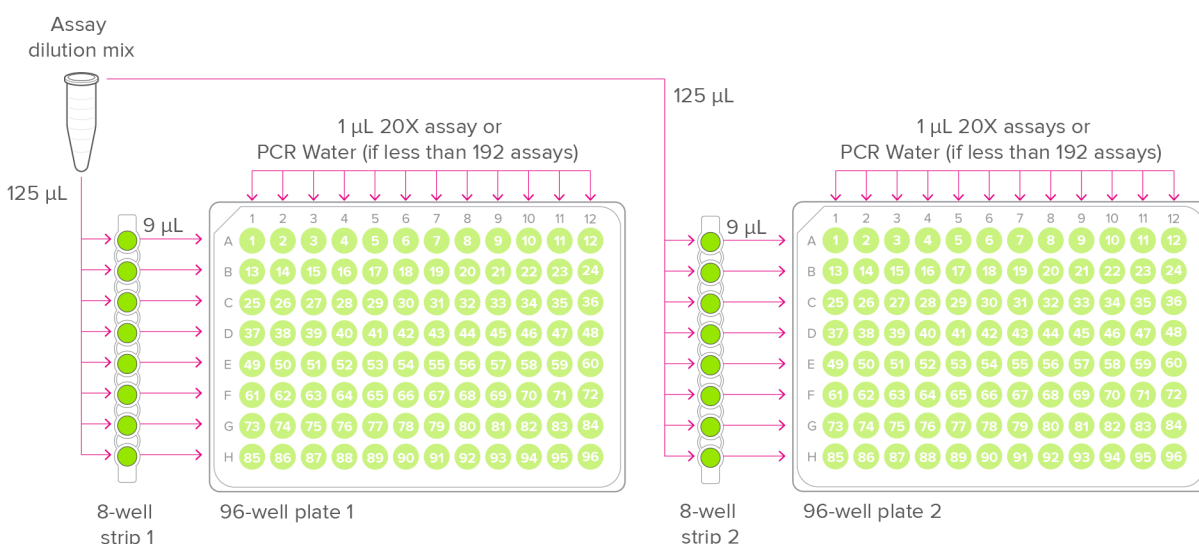


Figure 1. Preparation of 2X assay mix

NOTE 10 μL of 2X assay mix is sufficient for three GE 24.192 IFC runs.

STOPPING POINT The 2X assay mix can be stored at 4 $^{\circ}\text{C}$ if used within 1 week, or at -20°C for longer storage. Protect assay mix from light.

Prepare the Final Assay Mix

- 1 In a DNA-free hood, transfer 86 μL of 2X Gene Expression Master Mix into each well of an 8-well strip.
- 2 Prepare the final assay mix using the volumes in following [Table 7](#).

Table 7. Final assay mix

Component	Volume (μL)*
TaqMan Gene Expression Master Mix (2X) (Thermo Fisher Scientific 4369016)	3.0
2X assay mix (see page 10)	3.0
Total	6.0

- a Using an 8-channel pipette, transfer 3.0 μL master mix into each well of two new 96-well plates (see [Figure 2](#)).
- b Transfer 3.0 μL of 2X assay mix into each well of the two final assay mix plates.

IMPORTANT Wells in the assay plates that do not contain any assay must also include master mix in the final assay mix. All sample and assay inlets indicated on the IFC loading map must be filled with reagent for proper loading.

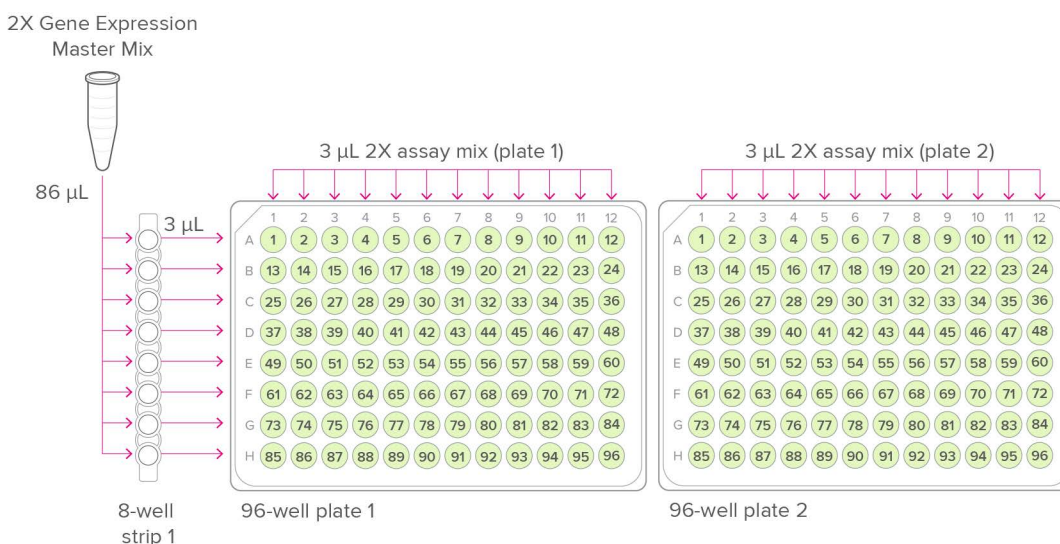


Figure 2. Preparation of final assay mixes

- 3 Seal the plates, vortex and centrifuge the reagents for 60 seconds.
- 4 The final assay mix contains 1X master mix, 1X assay mix and 1X GE Sample Loading Reagent. Set the plates aside until you are ready to load the IFC.

IMPORTANT Use the prepared assay mixes within 4 hours.

If you have not prepared the diluted samples, go to [Dilute the Preamplified cDNA and Prepare Final Sample Mixes](#).

Dilute the Preamplified cDNA and Prepare Final Sample Mixes

IMPORTANT The preamplified cDNA must be diluted to at least a 1:5 dilution to avoid primer carry-over into the real-time PCR reaction.

- 1 While the preamplification reactions are running, prepare the 1.25X sample dilution mix in a new 1.5 mL tube using volumes in [Table 8](#).

Table 8. Sample dilution mix

Components	Volume for 24 reactions (μL)
20X GE Sample Loading Reagent (Fluidigm PN 85000746)	35
DNA Suspension Buffer (Teknova PN T0223)	525
Total	560

- 2 After the preamplification reaction is finished, briefly vortex and centrifuge the preamplified sample mix plate and the reagents for 10 seconds.
- 3 Dilute the preamplification reaction 1:5 by adding 20 μL of sample dilution mix to each well of the preamplified sample mix plate. Set aside these final sample mixes until you are ready to load the GE 24.192 IFC (see [page 12](#)).

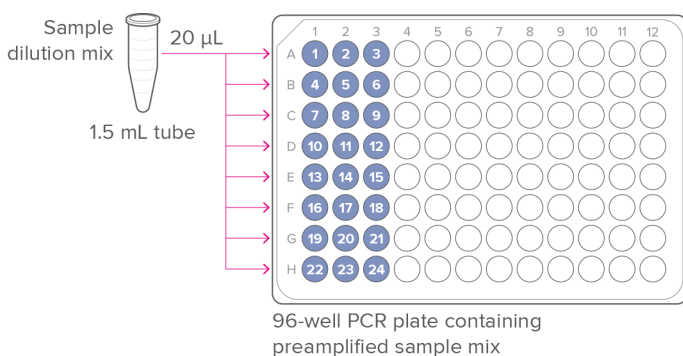


Figure 3. Preparation of final sample mixes

STOPPING POINT The diluted preamplification products can be used immediately. The sample can also be stored at 2–8 °C if used within 1 week, or stored at –20 °C for later use.

Load the IFC

Priming and loading on the GE 24.192 IFC occurs in one step on both Juno and the IFC Controller RX. Inject control line fluid and load other pressure fluids first, then load samples and assays.

For detailed instructions about injecting control line fluid, see the Control Line Fluid Loading Procedure (PN 68000132). For more information about using Juno, see the Juno System User Guide (PN 100-7070). For more information about using the IFC Controller RX, see the IFC Controller User Guide (PN 100-3385).

IMPORTANT

- Use the GE 24.192 IFC within 24 hours of opening the package.
- Only use GE 24.192 syringes with 150 μ L of control line fluid (Fluidigm PN 100-4058 or 101-7602).
- Do not evacuate air from syringes prior to injecting control line fluid.
- Avoid bending the syringe tip. Be careful when removing the syringe cap to prevent drips.
- Avoid getting Control Line Fluid on the exterior of the IFC or in the inlets because this makes the IFC unusable. If this occurs, use a new IFC.

- 1 Turn on the Biomark HD or Biomark system (computer and instrument), if necessary, and launch the Data Collection software to allow the instrument to start up and the camera to cool to the appropriate temperature.
- 2 Inject control line fluid into top accumulator on the IFC (see Figure 5). Use the entire contents of the syringe in the accumulator. Holding the IFC at a 45-degree angle, use the syringe tip to gently press down on the black O-ring to move it, insert the tip to one side of the accumulator, and then release the fluid (see Figure 4).

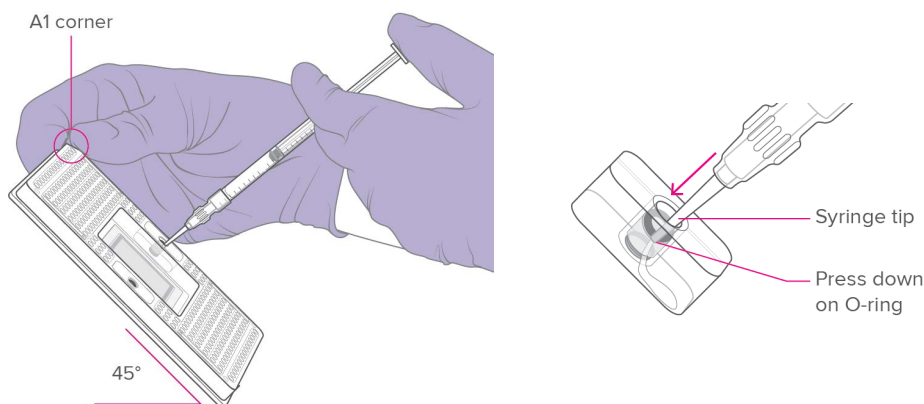


Figure 4. Injecting control line fluid into the accumulator

- 3 Remove and discard the protective film from the bottom of the IFC.

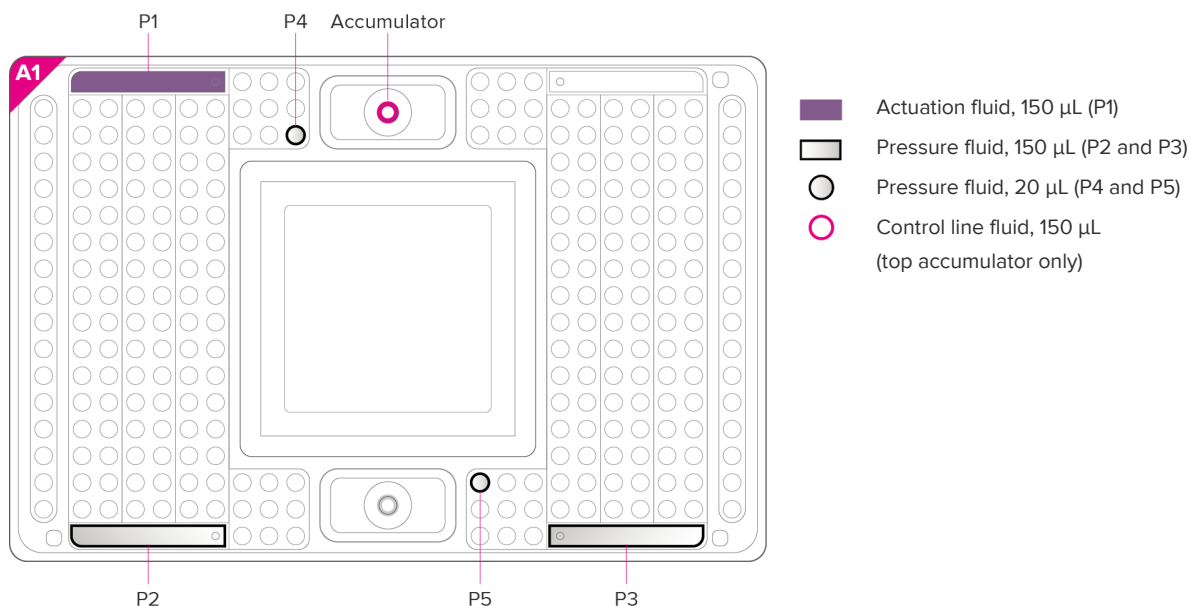


Figure 5. Priming map for the GE 24.192 IFC

- 4 Pipet 150 μL of Actuation Fluid into the P1 inlet (see Figure 5).
- 5 Pipet 150 μL of Pressure Fluid into the P2 and P3 reservoirs, respectively (see Figure 5). Ensure that the Pressure Fluid covers the reservoir bottom.

NOTE To evenly distribute reagents and avoid creating bubbles in reservoirs, pipet outward from edge closest to the center of the IFC and do not allow the pipette tip to touch the inlet hole



- 6 Using a P-20 pipette, load 20 μL of Pressure Fluid slowly into the P4 and P5 inlets, respectively (see Figure 5). Ensure that the fluid covers the inlet bottom.
- 7 Pipet the final assay mixes and sample mixes into the GE 24.192 IFC as shown in Figure 6 and Figure 7 on page 15.

IMPORTANT

- Vortex thoroughly and centrifuge all assay and sample final mix solutions before pipetting into the IFC inlets. Failure to do so may result in a decrease in data quality.
 - While pipetting, do not go past the first stop on the pipette. Doing so may introduce air bubbles into inlets.
 - Note the orientation of the A1 corner before you start pipetting.
- a Following the loading map in Figure 6, pipet 4 μL of each final assay mix into the 192 inlets of the IFC. Pipet one plate of 96 assays on the left side of the IFC, and another plate of assays on the right side

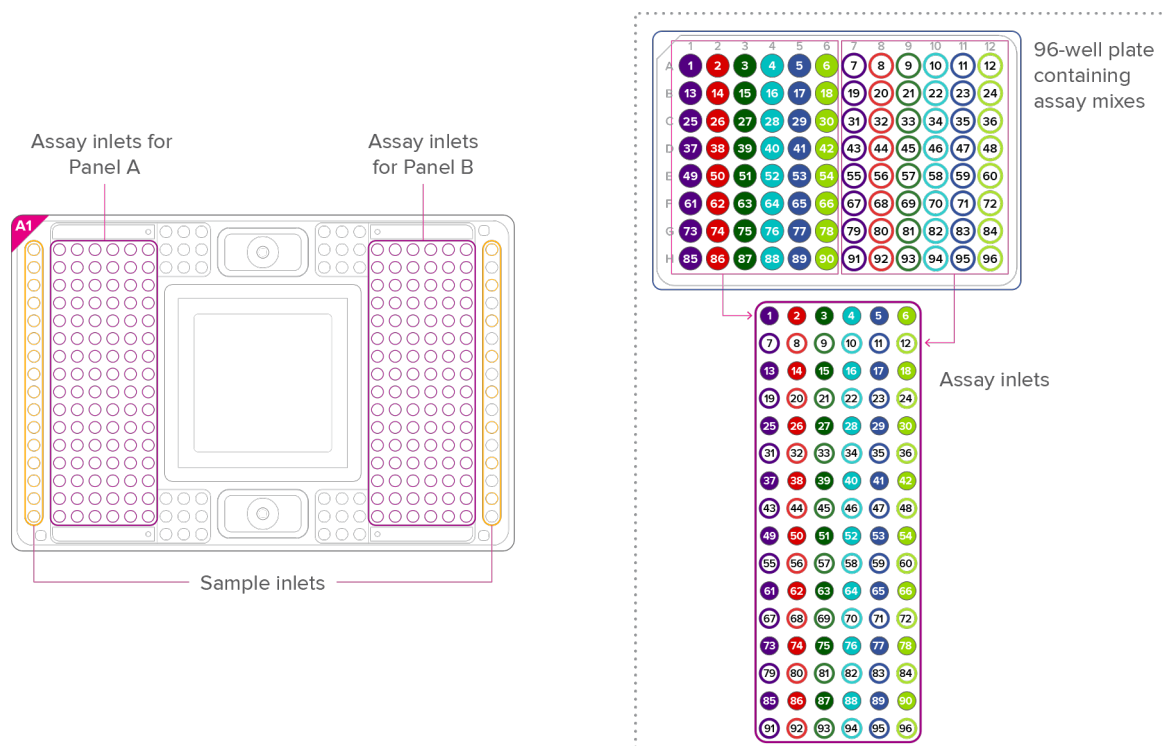


Figure 6. Assay loading map

- b** Following the loading map in Figure 7, load 4 μ L of sample final mix into the sample inlets.

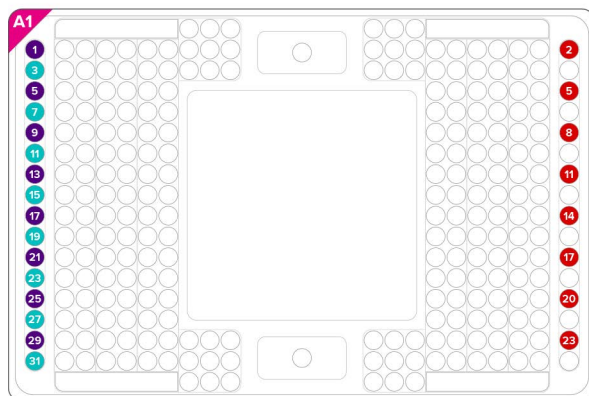
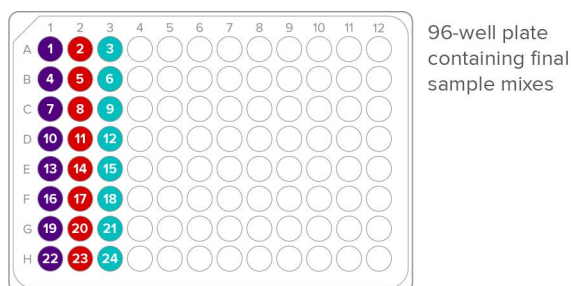


Figure 7. Sample loading map using an 8-channel pipette

- 8** Insert the IFC to an IFC controller or Juno, run the script:
- Juno: Tap **Load Mix 24.192 GE**, and then tap **Run**.
 - IFC Controller RX: Select **Load Mix (158x)** and press **Run Script**.

The script takes approximately 30 minutes to run.

- 9** After loading is complete, eject the IFC from the controller and collect real-time PCR data on Biomark HD or Biomark.

IMPORTANT Start the IFC run within 60 minutes of completing the loading script.

Collect Real-Time PCR Data

Transfer the loaded IFC from the controller to the Biomark HD or Biomark and collect real-time PCR data. For more information about using Biomark HD, see the Biomark HD Data Collection User Guide (PN 100-2451). For more information about using Biomark, see the Biomark/EP1 Data Collection User Guide (PN 68000127)

- 1** Use clear tape to remove any dust particles or debris from the IFC surface.
- 2** If necessary, double-click the **Data Collection** icon on the desktop of the Biomark HD system computer to launch the software.
- 3** Click **Start a New Run**.

- 4 Confirm that the camera status indicator (both Biomark HD and Biomark) and the lamp status indicator (Biomark only) at the bottom of the window are green.

Biomark HD:  Camera Temperature: -5.0 °C

Biomark:  Camera Temperature: -20.0 °C  Lamp is locked

- 5 Place the IFC on the instrument tray, aligning the notched A1 corner on the IFC with the A1 on the tray, and click **Load**.
- 6 Complete the Chip Barcode and Type section:
- Verify IFC barcode and IFC type.
 - Choose project settings (if applicable) and click **Next**.
- 7 Complete the Chip Run section by selecting either a new or pre-defined run.
- 8 Complete the Chip Run Name and Location section:
- Enter a run name or select the checkbox to use the IFC barcode as the run name.
 - Select a file storage location for a new IFC run or browse to select a pre-defined run file and click **Next**.
- 9 Complete the Application, Reference and Probes section and then click **Next**.

For...	Select
Application	Gene Expression
Passive reference	ROX
Assay	Single Probe
Probes	FAM-MGB

- 10 Complete the Thermal Protocol section:
- Browse to and select the thermal protocol: **GE 24x192 Standard v1.pcl**
For a description of the thermal protocol, see [Appendix C](#).
 - Confirm that **Auto Exposure** is selected and click **Next**.
- 11 Verify the IFC run information and click **Start Run**.
The IFC run takes approximately 1.5 hours.

Analyze the Data

After the run is complete, process your data using the Real-Time PCR Analysis software.

When analyzing data, we recommend you set the Ct Threshold Method to **Auto (Detectors)**, but you may need to manually adjust the quality threshold depending on the quality of your sample set (the default threshold is 0.65). For information about using the analysis software, see the Real-Time PCR Analysis User Guide (PN 68000088).

Appendix A: Available Bundle Kits

The 24.192 GE IFC kits come in 2-IFC and 10-IFC bundles. If you are using the Fluidigm protocol for cDNA preparation and preamplification, use the bundled kits that include the Preamp and Reverse Transcription Master Mix kit.

Bundle	Component	Part Number	Quantity
Biomark™ 24.192 Complete Bundle—Probe-Based GE—2 IFCs (PN 101-7659)	Biomark 24.192 GE Reagent Kit—2 IFCs	101-7589	1 kit
	Preamp and Reverse Transcription Master Mix —1 Tube	100-6300	1 kit
	24.192 Dynamic Array™ IFC for Gene Expression	101-7588	2 IFCs
	Biomark Control Line Fluid 192.24 & 24.192 (150 µL each)	101-7602	2 syringes

Bundle	Component	Part Number	Quantity
Biomark™ 24.192 Complete Bundle—Probe-Based GE—10 IFCs (PN 101-7660)	Biomark 24.192 GE Reagent Kit—10 IFCs	101-7591	1 kit
	Preamp and Reverse Transcription Master Mix Kit— 4 Tubes	101-7747	1 kit
	24.192 Dynamic Array™ IFC for Gene Expression	101-7588	10 IFCs
	Biomark Control Line Fluid 192.24 & 24.192 (150 µL each)	100-4058	10 syringes

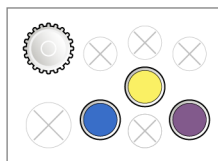
The following bundled kits are available if you are using your own protocol for cDNA preparation and preamplification.

Bundle	Component	Part Number	Quantity
Biomark™ 24.192 GE IFC & Reagent Kit—2 IFCs (PN 101-7690)	Biomark 24.192 GE Reagent Kit—2 IFCs	101-7589	1 kit
	24.192 Dynamic Array IFC for Gene Expression	101-7588	2 IFCs
	Biomark Control Line Fluid 192.24 & 24.192 (150 µL each)	101-7602	2 syringes

Bundle	Component	Part Number	Quantity
Biomark™ 24.192 GE IFC & Reagent Kit—10 IFCs (PN 101-7592)	Biomark 24.192 GE Reagent Kit—10 IFCs	101-7589	1 kit
	24.192 Dynamic Array™ IFC for Gene Expression	101-7588	10 IFCs
	Biomark Control Line Fluid 192.24 & 24.192 (150 µL each)	100-4058	10 syringes

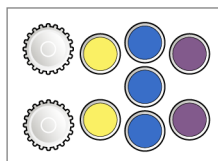
Appendix B: Reagent Kit Components

Biomark 24.192 GE Reagent Kit—2 IFCs (PN 101-7589)



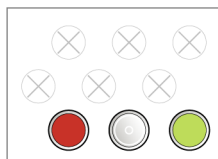
- 1 tube 20X GE Sample Loading Reagent, 1 mL (PN 85000746)
- 1 tube 2X Assay Loading Reagent, 1.5 mL (PN 85000736)
- 1 tube Actuation Fluid, 1.5 mL (PN 100-6250)
- 1 bottle Pressure Fluid, 2.04 mL (PN 100-6249)

Biomark 24.192 GE Reagent Kit—10 IFCs (PN 101-7591)



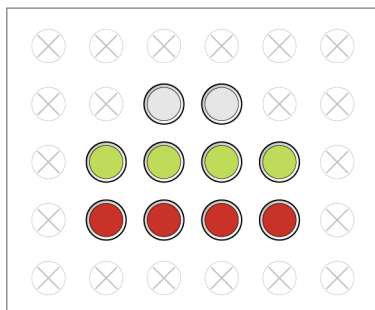
- 3 tubes 20X GE Sample Loading Reagent, 1 mL (PN 85000746)
- 2 tubes 2X Assay Loading Reagent, 1.5 mL (PN 85000736)
- 2 tubes Actuation Fluid, 1.5 mL (PN 100-6250)
- 2 bottles Pressure Fluid, 2.04 mL (PN 100-6249)

Preamp and Reverse Transcription Master Mix—1 Tube (PN 100-6300)



- 1 tube Reverse Transcription Master Mix, 106 μ L (PN 100-6297)
- 1 tube Preamp Master Mix, 106 μ L (PN 100-5744)
- 1 tube PCR Water, 1.8 mL (PN 100-5941)

Preamp and Reverse Transcription Master Mix—4 Tubes (PN 101-7747)



- 4 tubes Reverse Transcription Master Mix, 106 μ L (PN 100-6297)
- 4 tubes Preamp Master Mix, 106 μ L (PN 100-5744)
- 2 tubes PCR Water, 1.8 mL (PN 100-5941)

Appendix C: Biomark HD and Biomark Thermal Cycler Protocol

The GE 24.192 Standard v1 thermal cycling parameters are:

Temperature	Time	Cycles	Description
50 °C	2 min	1	UNG
95 °C	10 min		Hot start
95 °C	15 sec	40	Denaturation
60 °C	60 sec		Annealing

* Uracil DNA glycosylase

Appendix D: Related Documents

Go to fluidigm.com/documents to download these related documents.

Title	Part Number
Gene Expression Analysis with the GE 24.192 IFC Using TaqMan Assays Quick Reference	101-7088
Control Line Fluid Loading Procedure	68000132
IFC Controller RX User Guide	100-3385
Juno System User Guide	100-7070
Biomark HD Data Collection User Guide	100-2451
Biomark/EP1 Data Collection User Guide	68000127
Real-Time PCR Analysis User Guide	68000088

Appendix E: Safety

General Safety

In addition to your site-specific safety requirements, Fluidigm recommends the following general safety guidelines in all laboratory and manufacturing areas:

- Use personal protective equipment (PPE): safety glasses, fully enclosed shoes, lab coats, and gloves.
- Know the locations of all safety equipment (fire extinguishers, spill kits, eyewashes/showers, first-aid kits, safety data sheets, etc.), emergency exit locations, and emergency/injury reporting procedures.
- Do not eat, drink, or smoke in lab areas.
- Maintain clean work areas.
- Wash hands before leaving the lab.

Instrument Safety

For complete instrument safety information, including a full list of the symbols on the instrument, refer to the Juno System User Guide (PN 100-7070), IFC Controller RX User Guide (PN 100-3385), and Biomark HD Data Collection User Guide (PN 100-2451).



WARNING BIOHAZARD. If you are putting biohazardous material on the instrument, use appropriate personal protective equipment and adhere to Biosafety in Microbiological and Biomedical Laboratories (BMBL), a publication from the Centers for Disease Control and Prevention, and to your lab's safety protocol to limit biohazard risks. If biohazardous materials are used, properly label the equipment as a biohazard. For more information, see the BMBL guidelines online at: cdc.gov/biosafety/publications/index.htm.

Chemical Safety

The responsible individuals must take the necessary precautions to ensure that the surrounding workplace is safe and that instrument operators are not exposed to hazardous levels of toxic substances. When working with any chemicals, refer to the applicable safety data sheets (SDSs) provided by the manufacturer or supplier.

Disposal of Products

Used IFCs and reagents should be handled and disposed of in accordance with federal, state, regional, and local laws for hazardous waste management and disposal.

For technical support visit fluidigm.com/support.

North America +1 650 266 6100 | Toll-free (US/CAN): 866 358 4354 | techsupport@fluidigm.com Latin America +1 650 266 6100 | techsupportlatam@fluidigm.com
Europe/Middle East/Africa/Russia +44 1223 859941 | techsupporteurope@fluidigm.com China (excluding Hong Kong) +86 21 3255 8368 | techsupportchina@fluidigm.com
Japan +81 3 3662 2150 | techsupportjapan@fluidigm.com All other Asian countries/India/Australia +1 650 266 6100 | techsupportasia@fluidigm.com

For Research Use Only. Not for use in diagnostic procedures.

Limited Use License to Perform Preamplification with Fluidigm IFCs: A license to use Thermo Fisher Scientific's patented preamplification method workflows involving a Fluidigm integrated fluidic circuit (IFC) can be obtained (i) with purchase of a Fluidigm IFC from Fluidigm Corporation or (ii) by a separate license from Thermo Fisher Scientific. For licensing information, contact outlicensing@lifetech.com.

NOTICE TO PURCHASER: LIMITED LICENSE: Use of this product is covered by one or more US patents. The purchase of this product includes a limited, non-transferable immunity from suit under the foregoing patent claims for using only this amount of product for the purchaser's own internal research. No right under any other patent claim, no right to perform any patented method, and no right to perform commercial services of any kind, including without limitation reporting the results of purchaser's activities for a fee or other commercial consideration, is conveyed expressly, by implication, or by estoppel. This product is for research use only. Diagnostic uses under Roche patents require a separate license from Roche. Further information on purchasing licenses may be obtained by contacting outlicensing@lifetech.com or Out Licensing, Thermo Fisher Scientific, 5781 Van Allen Way, Carlsbad, California 92008.

Information in this publication is subject to change without notice. **Patent and license information:** fluidigm.com/legal/notice. Fluidigm, the Fluidigm logo, Biomark, Dynamic Array, EP1, and Juno are trademarks and/or registered trademarks of Fluidigm Corporation in the United States and/or other countries. All other trademarks are the sole property of their respective owners. © 2018 Fluidigm Corporation. All rights reserved. 05/2018